

Visual Elements and Graphics

Chapter-5

Configuring Lines and Borders

- horizontal rule element
- border and padding properties

horizontal rule element <hr>

- visually separates areas of a page
- configures a horizontal line across a web page
- a void tag (A void tag (also called a self-closing tag) is an HTML element that cannot have any content or closing tag — it stands alone)

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Horizontal Line Example</title>
<style>
  hr{ border: 5px solid blue; border-radius: 2px;}
  h2{text-align: center;}
</style>
</head>
<body>
  <hr>
  <h2>Welcome to Mandalay Technological University </h2>
  <hr>
</body></html>
```



CSS Border Properties

Property	usage	example	
<code>border-color</code>	The color of the border around an element	<code>border-color:#00FF00;</code>	
<code>border-left</code>	configure the left border of an element	<code>border-left: 1px solid #000000;</code>	<code>border-width, border-style, border-color</code>
<code>border-right</code>	configure the right border of an element	<code>border-right: 1px solid #000000;</code>	
<code>border-top</code>	configure the top border of an element	<code>border-top: 1px solid #000000;</code>	
<code>border-bottom</code>	configure the bottom border of an element	<code>border-bottom: 1px solid #000000;</code>	
<code>border-width</code>	The width of a border	<code>border-width: thin;</code> <code>border-width: 5px;</code>	
<code>border-style</code>	The type of border around an element	<code>border-style: double;</code>	<code>double, groove, inset, none (the default), outset, ridge, solid, dashed, dotted, and hidden</code>
<code>border-radius</code>	Configures rounded corners on a element.	<code>border-radius: 15px;</code>	

Sample Code 1

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Dashed Border Example</title>
    <style>
      .dashedborder{
        border-width: 3px;
        border-style: dashed;
        border-color: #2fbf2a;
        text-align: center;
      }
    </style>
  </head>
```

```
<body>
  <h1>Example of Dashed Border</h1>
  <div class="dashedborder">
    This box has a dashed border!
  </div>
</body>
</html>
```

Sample Code 2

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>CSS Border Properties Example</title>
<style>
.style-box {
border-color: #3498db;
border-left: 5px solid #2ecc71;
border-right: 3px dashed #e74c3c;
border-top: 4px dotted #9b59b6;
border-bottom: 2px double #f1c40f;
border-width: 15px;
border-style: solid;
border-radius: 15px;
text-align: center;
font-weight: bold;
background-color: #67a8b9d4; }
```

```
</style>
</head>
<body>
<h1>Example of Border Properties</h1>
<div class="style-box">
This box is an example of multiple border
properties.
</div>
</body>
</html>
```

CSS Padding Properties

Property	Usage	Example	remark
padding	configure the amount of padding—the blank space between the element and its border	padding: 10px 20px 30px 40px; padding:20px;	Top, right, bottom, left
padding-bottom	Blank space between an element and its left border	padding-bottom: 10px	
padding-left	Blank space between an element and its left border	padding-left:20px	
padding-right	Blank space between an element and its right border	padding-right:30px	
padding-top	Blank space between an element and its top border	padding-top:20px	

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title> Padding Example</title>
  <style>
    .padding-box {
      width: 200px;
      height: 100px;
      background-color: lightblue;
      border: 2px solid blue;
      padding: 30px 40px 25px 15px;
    }
  </style>
</head>
```

```
<body>
  <h1>Unified Padding Example</h1>
  <div class="padding-box">
    This box uses padding
    -Top: 30px
    -Right: 40px
    -Bottom: 25px
    -Left: 15px
  </div>
</body>
</html>
```

Sample Code

Unified Padding Example

This box uses padding - Top:
30px - Right: 40px - Bottom:
25px - Left: 15px

CSS Background Properties

Property	Description	Example	Remark
<code>background-attachment</code>	Configures whether the background image scrolls with the page or is fixed in place	<code>background-attachment: fixed;</code>	<code>fixed</code> , <code>scroll</code> (default)
<code>background-image</code>	Background image on an element	<code>background-image: url("paper.jpg");</code>	To display an image, use <code>url(imagename.gif/jpg/png)</code> . To disable image display, use <code>none</code> (default).
<code>background-position</code>	Position of the background image	<code>background-position: center;</code>	<code>repeat</code> <code>repeat-x</code> <code>repeat-y</code> <code>no-repeat</code>
<code>background-repeat</code>	Controls how the background image will repeat	<code>background-size: auto;</code>	<code>auto</code> <code>length</code> <code>cover</code> <code>contain</code>
<code>background-size</code>	Configures the size of the background image. This CSS3 property is not supported in all browsers.	<code>background-size: cover;</code>	<code>auto</code> <code>length</code> <code>cover</code> <code>contain</code>

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Background Properties Example</title>
  <style>
    .background-example {
      width: 100%;
      height: 600px;
      background-image: url('beach.jpg');
      background-attachment: fixed;
      background-position: center;
      background-repeat: no-repeat;
      background-size: cover;
    }
  </style>
```

```
</head>
<body>
  <div class="background-example">
    <h1 style="color: Blue; text-align: center;
padding-top: 100px;">Background Properties
Example</h1>
  </div>
</body>
</html>
```

Output...



Types of Graphics

- Graphic Interchange Format (GIF) Image
- Joint Photographic Experts Group (JPEG) Images
- Portable Network Graphic (PNG) Images

Image Element

- The image element configures graphics on a web page.
- These graphics can be photographs, banners, company logos, navigation buttons, and so on.

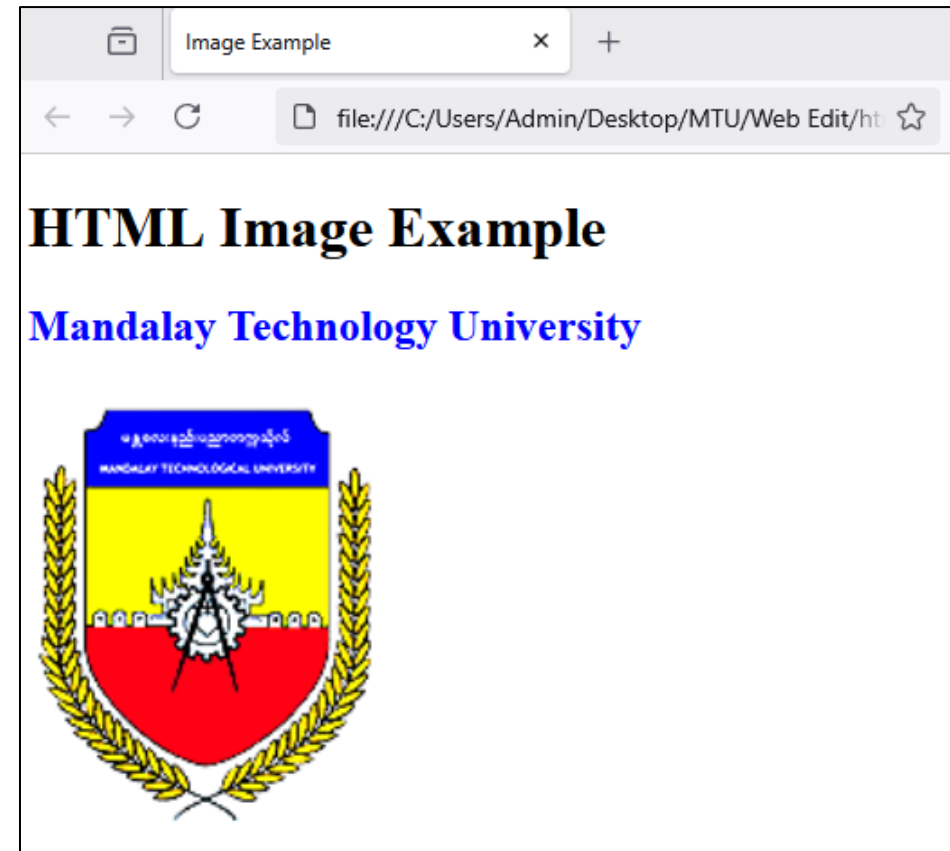
```

```

- src-The URL or file name of the image
- alt -Text phrase that describes the image
- height -Height of image in pixels
- width -Width of image in pixels

Sample Code

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <title>Image Example</title>
  </head>
  <body>
    <h1>HTML Image Example</h1>
    <h2 style="color: blue;">Mandalay
    Technology University</h2>
    
  </body>
</html>
```

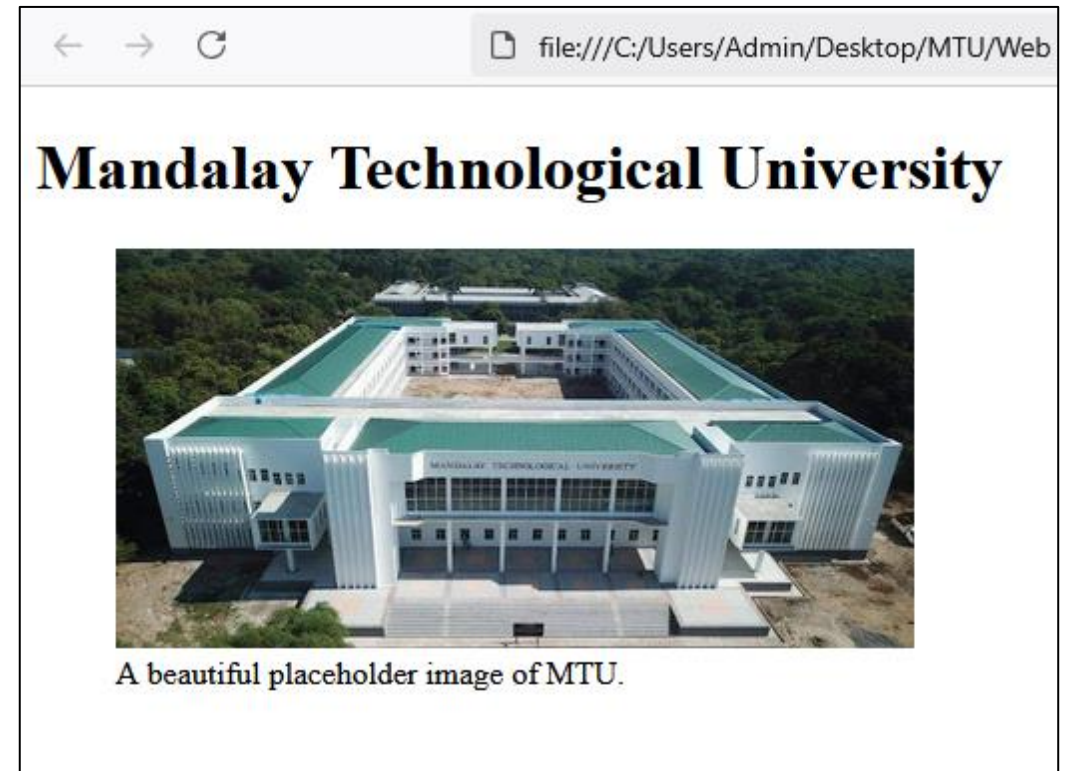


HTML5 Visual Elements

- figure element -a unit of content that is self-contained, such as an image, along with one optional figcaption element
- figcaption element -provides a caption for a figure
- meter element -displays a visual gauge of a numeric value within a known range
- progress element -displays a bar that depicts a numeric value within a specified range.

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Figure and Figcaption Example</title>
</head>
<body>
<h1>Mandalay Technology University</h1>
<figure>
<imgsrc="mtuphoto.jpg" alt="MTU"
width="400" height="200">
<figcaption>A beautiful placeholder image of
MTU.</figcaption>
</figure>
</body></html>
```



Sources of Graphics

- Free Images: <http://www.freeimages.com>
- Free Stock Photo Search Engine: <http://www.everystockphoto.com>
- Free Digital Photos: <http://www.freedigitalphotos.net>
- The Stock Solution: <http://www.tssphoto.com>
- Pixabay: <http://pixabay.com>
- iStockphoto: <http://www.istockphoto.com>

Creating banners and button images

- LogoCreator: <http://creatr.cc/creatr>
- Cool Text <http://www.cooltext.com>
- Da Button Factory: <http://dabuttonfactory.com>

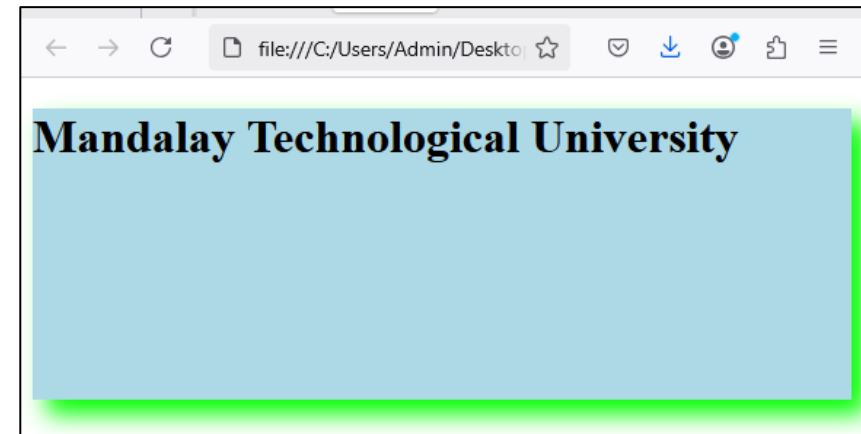
CSS3 box-shadow property

- to create a shadow effect on block-display elements such as div and paragraph elements.
- **Horizontal Offset**: Use a number. Positive values move the shadow to the right; negative values move it to the left.
- **Vertical Offset**: Use a number. Positive values place the shadow below; negative values place it above.
- **Blur Radius** (optional): Use a number. Defaults to 0 for sharp shadows. Higher values make the shadow blurrier.
- **Spread Distance** (optional): Use a number. Defaults to 0. Positive values make the shadow larger; negative values make it smaller.
- **Color**: Use a valid color for the shadow.
- `box-shadow: 5px 5px 5px #828282;`

CSS3 box-shadow property

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Box Shadow Example</title>
<style>
.box {
width: 100%;
height: 200px;
background-color: lightblue;
box-shadow: 10px 10px 15px rgba(0, 0, 0, 0.3);
}
</style>
</head>
```

```
<body>
  <div class="box">
    <h1 class="box2">Mandalay
    Technological University</h1>
  </div>
</body>
</html>
```



CSS3 opacity property

- opacity property configures the transparency of an element.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Opacity Property
Example</title>
  <style>
    #content { background-image:
url(NSPU\ Uni.jpg);

    margin-left: auto;
    margin-right: auto;
    width: 100%;
    height: 900px;
    padding-top: 20px;}
```

```
h1 { background-color: #FFFFFF;
color: rgba(10,10,255);
opacity: 0.5;
font-size: 18 px;
padding: 30px;
margin-top: 200px;
text-align: center;
}

</style>
</head>
<body>
  <div id="content">
    <h1 >Naypyitaw State Polytechnic
University</h1>
  </div></body></html>
```


CSS3 opacity property



CSS3 RGBA color

- RGBA means Red, Green, Blue, and Alpha
- Alpha: controls transparency (opacity).
- `rgba(0,255,0, 1)` = Fully opaque green.
- `rgba(0,255, 0, 0.5)` = 50% transparent green.
- `rgba(0, 255, 0, 0)` = Fully transparent (invisible) green.

CSS3 Gradients

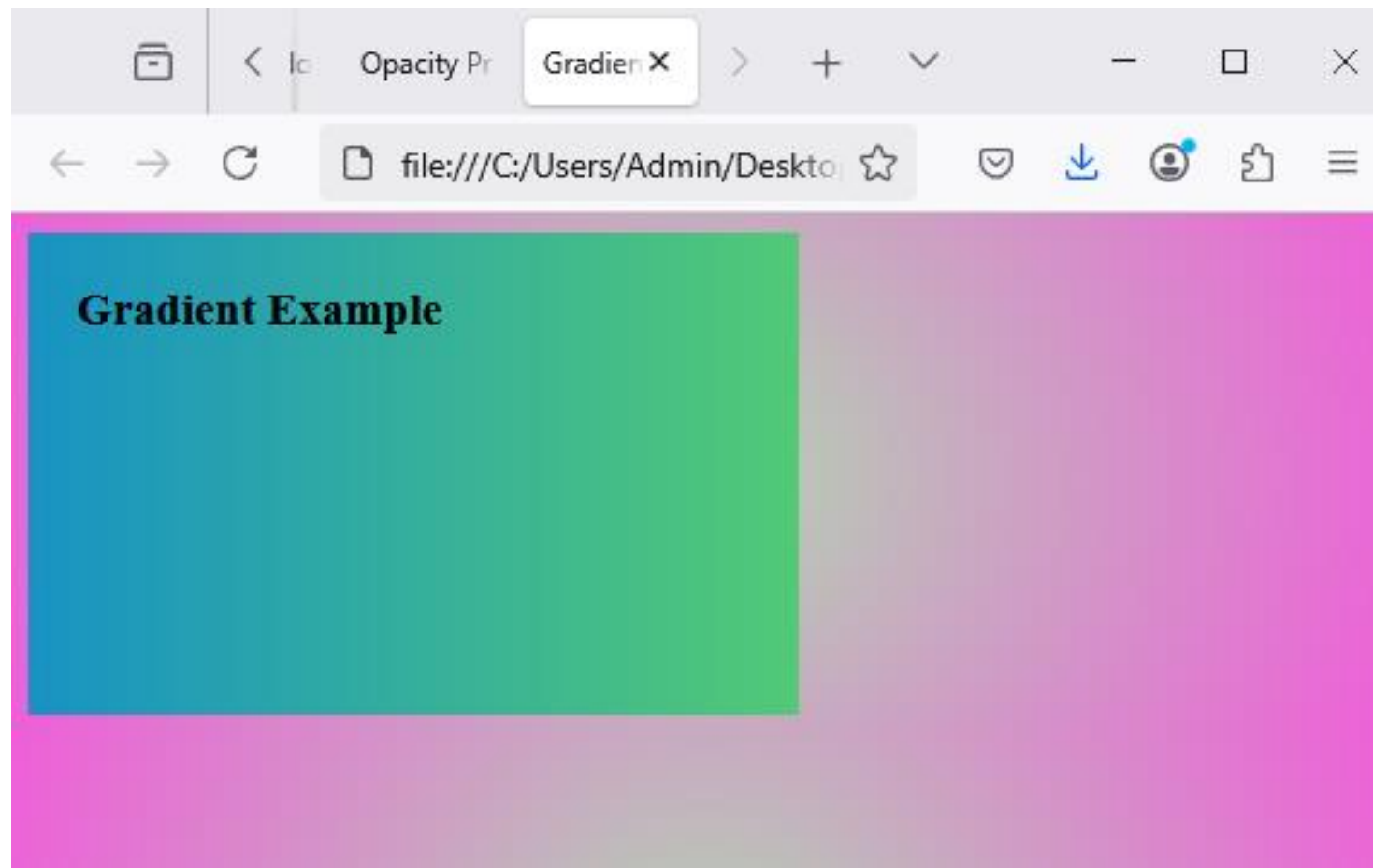
- **linear gradient** : a smooth blending of color in a single direction such as from top to bottom or from left to right.
 - `background-image: linear-gradient(to bottom, #FFFFFF, #00FF00);`
- **radial gradient** : a smooth blending of color emanating outward from a single point
 - `background-image: radial-gradient(#FFFFFF, #0000FF);`

Sample Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title> Gradient Example</title>
<style>
.linear-gradient {
width: 300px;
height: 180px;
background: linear-gradient(to right,
rgb(24, 147, 196), rgb(82, 202, 118));
font-size: larger;
padding-top: 20px;
font-weight: bold;
padding-left: 20px;}
```

```
body
{
background-image: radial-gradient(circle,
#b3d4b3, #ef5ed9);
}
</style>
</head>
<body>
<div class="linear-gradient">Gradient
Example</div>
</body>
</html>
```

Output



Next Lecture

Web Design